

Ramya Ajwalia

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PROFESSIONAL SUMMARY

Systems Software Engineer (M.S. CS, GPA: 3.89) with expertise in C/C++, Java, JavaScript/TypeScript and Python; builds scalable, low-latency backend systems on Linux. Strong in concurrency, networking, and distributed systems, with experience designing microservices, improving performance, and reducing infrastructure costs while maintaining high availability.

PROFESSIONAL EXPERIENCE

AI Product Engineering Intern

Aug. 2025 – Aug. 2026

Quak Travel – Boston, USA

- Integrated AI services with FastAPI, boosting accuracy by 15%, reducing inference latency by 20%, and saving around \$20K/year in cloud costs.
- Engineered and optimized scalable backend services, reducing latency by 40% and improving throughput to 1.5K+ requests/sec in a distributed microservices architecture.
- Implemented concurrent processing and distributed APIs, cutting inter-service latency by approximately 30% and improving system scalability.
- Improved reliability via logging/monitoring, achieving 99.9%+ uptime and reducing downtime by 25%.

Web Developer

Jan. 2023 – Jun. 2024

Riya Softweb Solutions – India

- Developed and maintained Java-based CRM and HRMS platforms, building new features and resolving 50+ issues/month; cut resolution time 25–30% and improved user satisfaction 15–20%.
- Managed roles & DB updates; improved data integrity 20%, ensured system continuity.
- Documented Java workflows/fixes; reduced recurring issues approximately by 25%, improved team efficiency.
- Performed functional and REST API testing to validate application workflows, ensure data integrity, and identify defects; collaborated with developers to resolve issues and improve overall system reliability.

PROJECTS

Driver Drowsiness & Distraction Detection System

Oct. 2025

- Built real-time ADAS using MediaPipe Face Mesh; achieved <33ms/frame latency on CPU-only systems.
- Enabled 5–15% insurance cost reduction ($\approx$ \$50K–\$150K/year) through improved driver safety.
- Developed EAR, MAR, and head pose state machine; improved response time by 0.7s and safety score by 8 points.
- Reduced near-misses and vehicle downtime, improving overall fleet efficiency and operational reliability.

PrismEDU - Full-Stack Learning Management System

Jul. 2025

- Developed React frontend and Node.js/Spring Boot APIs with role-based access and scalable architecture.
- Optimized backend and database queries, reducing API response time by $\sim$ 25%.
- Tech Stack: React, Node.js, Express, PostgreSQL/MySQL, Python (Flask/FastAPI), AWS, GitHub.
- Performed UI automation using Selenium and REST API testing to validate functionality and integrations.

TECHNICAL SKILLS

- **Systems Programming:** C, C++, Python, Java, JavaScript, TypeScript, SQL, Object-Oriented Programming, Data Structures & Algorithms
- **Operating Systems & Linux:** Linux environment, Bash scripting, Process management, Memory management concepts, Multithreading, Concurrency fundamentals
- **Networking:** TCP/IP fundamentals, HTTP, Socket programming concepts, Distributed systems communication patterns
- **Testing Automation:** Selenium WebDriver, REST API Testing, Test Case Design, Functional & Integration Testing, Defect Identification, Debugging
- **Cloud & Distributed Systems:** AWS (EC2, S3, RDS), Microservices architecture, Event-driven systems, Fault tolerance
- **Debugging & Performance:** Logging, Exception handling, Profiling concepts, Performance tuning, Root cause analysis
- **DevOps & Tools:** Git (branching, merging), CI/CD fundamentals, Unit testing

EDUCATION

University of Texas at Arlington – Arlington, TX

Aug. 2024 – May. 2026

Master of Science in Computer and Information Science | CGPA: 3.90/4

Coursework: Database Administration, Data Analysis and Algorithms, Data Modeling Techniques, Software Engineering, Machine Learning, Cloud Computing and Big Data, Artificial Intelligence, Software Testing and Quality Assurance, Advance Topics in Software Engineering, Data Mining, Computer Vision, Computer Networks.

Gujarat Technological University – India

Jul. 2020 – Jul. 2024

Bachelor of Engineering in Information Technology | CGPA: 9.17/10